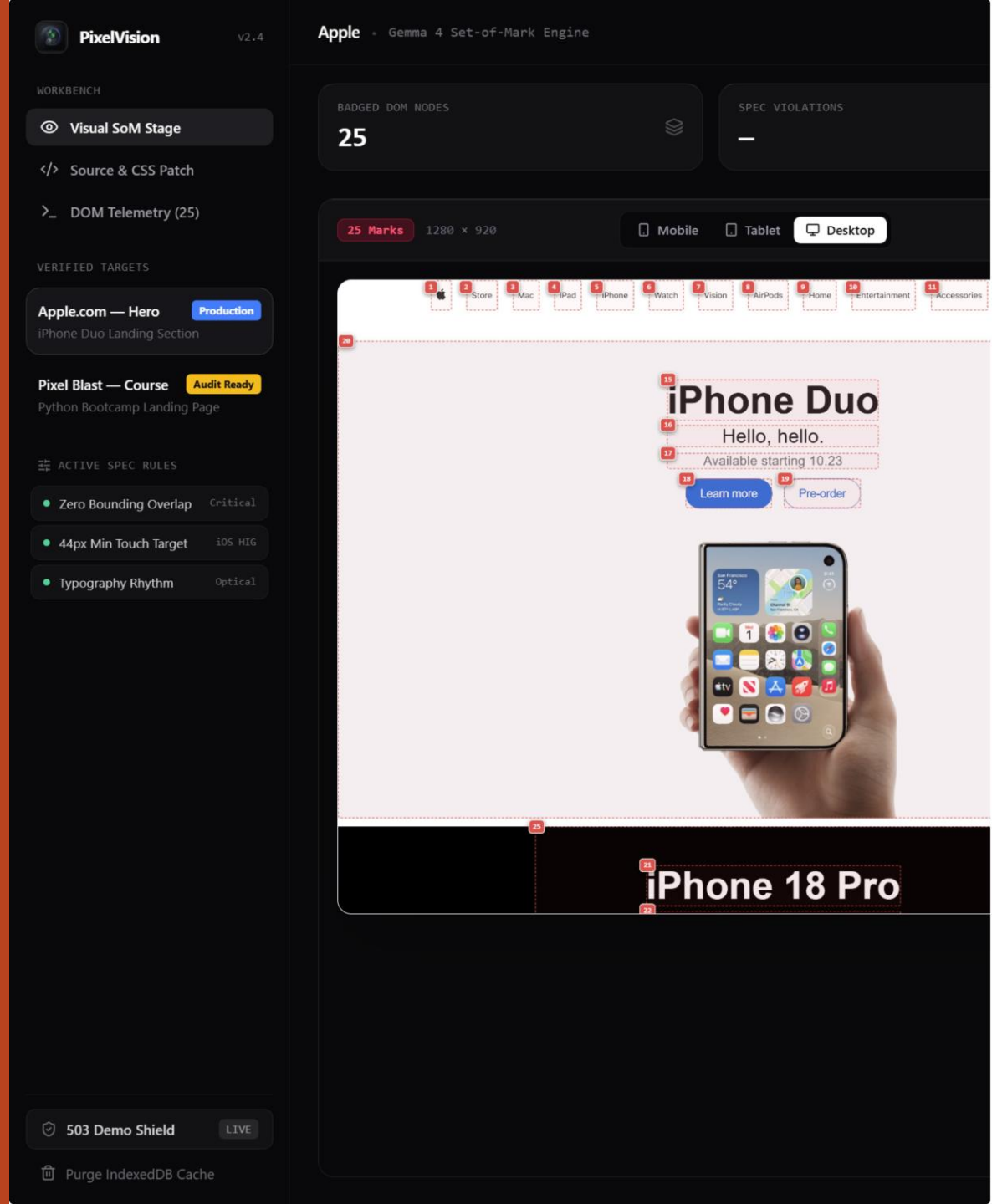
Track 01: Best Use of Gemma 4 · Problem Statement G3

PIXELVISION WORKBENCH

The Missing Visual Compiler for Frontend Engineering

Automated UI/UX auditing and self-healing workbench powered by Google's Gemma 4 (31B/26B) multimodal reasoning.

pixelvision-black.vercel.app



⚠ The Blindspot in CI/CD

Code tests are visually blind

Traditional test suites assert code existence, while user-facing design breaks in silence.

Green Terminal: Tests Passed

PASS 100%

DOM tests lack eyes

Jest and Cypress assert that buttons exist in the DOM tree, giving false confidence even when floating headers completely obscure them.

Broken Rendered UI: Fails Reality

FAIL HIG

Humans cannot eyeball math

Manual QA misses a 26px button violating Apple's strict 44px touch target rule. Design critiques stall in subjective debate instead of objective geometric verification.

Core Technology

Giving Gemma 4 spatial vision

Bridging raw DOM code and visual reality with grounded Set-of-Mark (SoM) indexing.

01

1. SoM Visual Tagging

Live iframe overlays inject numbered red bounding-box badges (#1 to #40) directly onto interactive DOM elements in real time.

02

2. Dual-Modality Payload

PixelVision bundles a Base64 rendered screenshot alongside an exact JSON coordinate matrix mapping X, Y, Width, and Height for every indexed node.

03

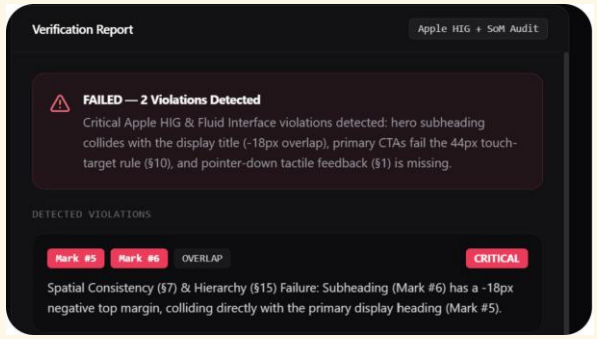
3. Objective HIG Reasoning

Gemma 4 evaluates the paired inputs against Apple HIG standards: §1 Pointer-down, §7 Spatial Consistency, and §10 44px Touch Targets.

End-to-End Remediation

From broken layout to self-healed CSS

One-click visual verification and automated remediation in seconds.



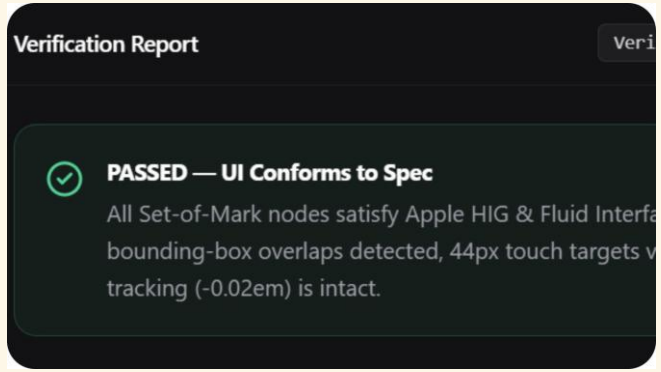
1. Harvest & Tag

Load Apple.com or Pixel Blast into the Visual SoM Stage and index 25+ live DOM nodes with bounding marks.



2. Simulate & Audit

Trigger a simulated z-index collision and touch-target defect; Gemma 4 isolates failing Marks #5 and #6.



3. 1-Click Self-Healing

Gemma 4 outputs a verified CSS patch, instantly injected into the runtime stage to restore HIG compliance.

System Design

Production-Grade Architecture

Engineered for resilience, speed, and zero-crash demos

Frontend

React 18, TypeScript (Strict), Vite, Tailwind CSS with Apple WWDC Fluid Interface styling

3-tier CORS capture

html-to-image with font-skipping, cross-origin media filtering, native Canvas synthetic SoM fallback

Zero-latency caching

SHA-256 hashing on IndexedDB (localforage) plus a 503 Demo Shield fallback

Iframe Sandbox

3-Tier Capture + SHA-256 Cache

Gemma 4 API

Hot-Reload Patch Engine

Impact & Delivery

From human guesswork to automated math

Framework agnostic verification built solo by a 16-year-old developer.

0ms

Cached Audit

Sub-millisecond replay via SHA-256 IndexedDB cache, eliminating model latency for re-inspections.

44px

Apple Design Compliance

Exact geometric touch-target guarantees across React, Vue, Svelte, and plain HTML applications.

1-Click

CSS Hot Patch

Automated diff generation injected directly into the live DOM without restarting builds.

Built solo by Goutham Kumar A · Class 11 student & indie developer of Pixel Focus on the Microsoft Store.

Test Live Demo